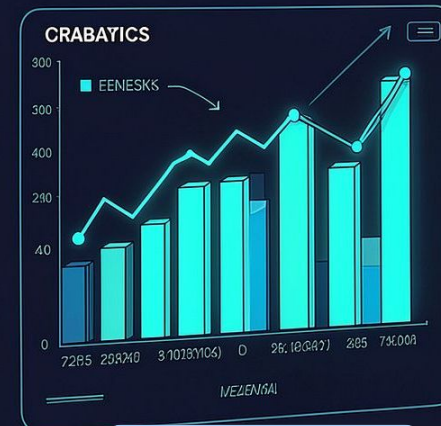


Customer Shopping Behavior Analysis

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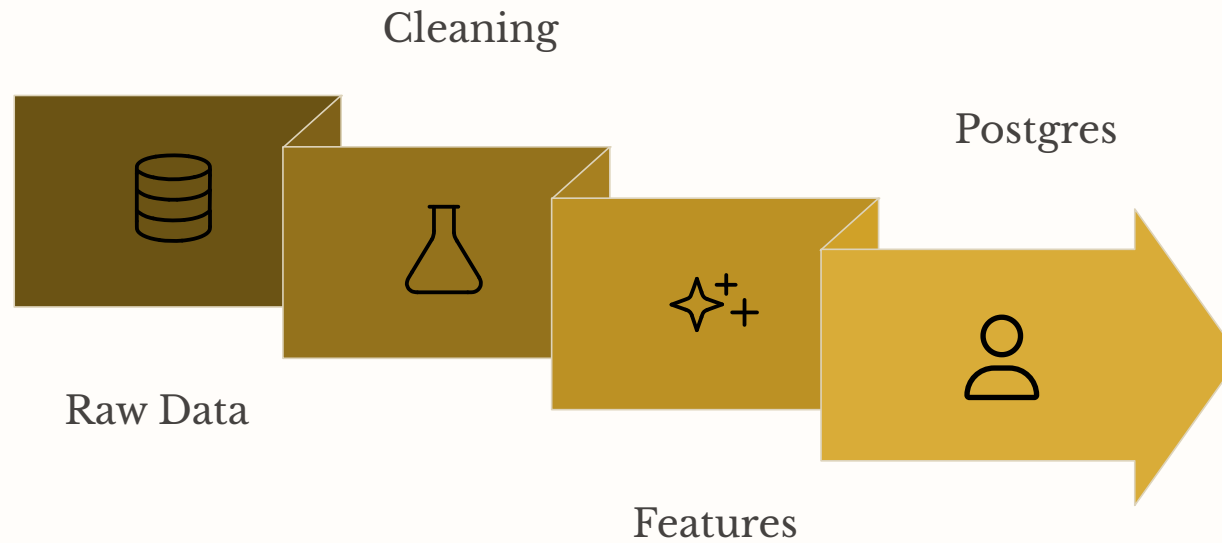
POWER BI



Made with **GAMMA**

Project Overview

This project transforms raw customer transaction data into meaningful business insights through an end-to-end data analytics workflow. It analyzes customer purchasing patterns, revenue performance, product ratings, subscription behavior, and repeat-purchase activity — all presented through an interactive Power BI dashboard.



The analysis focuses on six key areas: **customer behavior**, **revenue**, **product performance**, **subscription behavior**, **repeat purchases**, and **customer segmentation**.

Dataset Overview

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Visa
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	MasterCard
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Visa
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	MasterCard
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Visa

Dataset Summary

3,900

Customer purchase records

18

Columns / variables

Data Categories

- **Customer Demographics:** Age, Gender, Location
- **Product Info:** Item, Category, Size, Color, Season
- **Purchase Details:** Purchase Amount (USD)
- **Shopping Behavior:** Previous Purchases, Frequency
- **Subscription & Shipping:** Status, Shipping Type
- **Payment & Feedback:** Payment Method, Review Rating

Data Cleaning Using Python & Pandas

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Customer ID           3900 non-null  int64  
1   Age                   3900 non-null  int64  
2   Gender                3900 non-null  object  
3   Item Purchased        3900 non-null  object  
4   Category              3900 non-null  object  
5   Purchase Amount (USD) 3900 non-null  int64  
6   Location              3900 non-null  object  
7   Size                  3900 non-null  object  
8   Color                 3900 non-null  object  
9   Season                3900 non-null  object  
10  Review Rating         3863 non-null  float64 
11  Subscription Status   3900 non-null  object  
12  Shipping Type         3900 non-null  object  
13  Discount Applied      3900 non-null  object  
14  Promo Code Used       3900 non-null  object  
15  Previous Purchases    3900 non-null  int64  
16  Payment Method        3900 non-null  object  
17  Frequency of Purchases 3900 non-null  object  
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

Cleaning Steps & Findings

01

Data Loading & Exploration

Loaded 3,900 records across 18 columns. `df.info()` confirmed structure; Review Rating had 3,863 non-null values.

02

Missing Value Treatment

37 missing values in Review Rating handled via **category-level median imputation**, preserving records with category-specific values.

03

Standardization & Feature Engineering

Columns converted to `snake_case`. Redundant *Promo Code Used* column removed. New **Age** and **Frequency Mapping** features created for segmentation.

PostgreSQL Integration & SQL Business Questions

	customer_id bigint	age bigint	gender text	item_purchased text	category text	purchase_amount bigint	location text	size text	color text	season text	review_rating double precision	subscription_status text	shipping_ text
1	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express
2	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express
3	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Ship
4	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day
5	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquo...	Spring	2.7	Yes	Free Ship
6	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard
7	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Ship
8	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Ship
9	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express
10	10	57	Male	Handbag	Accessori...	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shi

Database Integration

Cleaned dataset loaded into **customer** table within `customer_behavior` database in PostgreSQL. Ten business questions were developed to analyze revenue, spending, product ratings, subscription behavior, discounts, loyalty, repeat purchases, and age-group contribution.

Q1

Total revenue: Male vs. Female customers

Q2

Customers using discounts but spending above average

Q3

Top 5 products by average review rating

Q4

Avg. purchase: Standard vs. Express shipping

Q5

Do subscribed customers spend more?

Q6

Top 5 products by discount rate

Q7

Segment: New, Returning, Loyal customers

Q8

Top 3 products per category

Q9

Repeat buyers (5+ purchases) subscription rate

Q10

Revenue contribution by age group

SQL Analysis: Key Visuals

Top 5 Products by Average Review Rating (Q3)

	item_purchased text	average_product_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Spending Comparison by Subscription Status (Q5)

	subscription_status text	total_customer bigint	average_revenue numeric	total_revenue numeric
1	Yes	1053	59.49	62645
2	No	2847	59.87	170436

Customer Segmentation by Previous Purchases (Q7)

	customer_segment text	number_of_customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

Repeat Buyers by Subscription Status (Q9)

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Revenue Contribution by Age Group (Q10)

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Total purchase revenue was aggregated for each age group and ranked from highest to lowest. The **Young-Adult** segment emerged as the top-performing age group in terms of revenue contribution, highlighting this demographic as a critical focus area for targeted marketing and personalized promotional strategies.

Interactive Power BI Dashboard



Dashboard Components

→ KPI Cards

Total Customers, Total Revenue, Average Purchase, Average Rating

→ Customer Behavior Visuals

Subscription Mix, Repeat Buyers by Subscription

→ Product & Revenue Analysis

Top 5 Products by Rating, Revenue by Age Group

→ Interactive Features

Slicers, dynamic insights, and Reset Filters button

Key Business Insights

~4K

Total Customers

\$59.76

Avg. Purchase

~3K

Repeat Buyers

\$233K

Total Revenue

3.75

Avg. Rating

27%

Subscribers

Top Findings

- **Young-Adult** segment leads revenue contribution
- **Boots** is the highest-rated product
- 73% of customers are non-subscribers — significant growth opportunity
- Strong repeat-buyer base (~3K) indicates recurring purchasing activity

Opportunities Identified

- Increase subscription adoption through targeted benefits
- Strengthen loyalty programs for repeat buyers
- Focus marketing on Young-Adult segment
- Promote high-rated products like Boots
- Improve lower-rated products to raise overall satisfaction

Recommendations & Future Scope

Business Recommendations

1 Increase Subscription Adoption

Exclusive discounts, loyalty rewards, and personalized offers for non-subscribers

2 Strengthen Repeat Customer Retention

Loyalty programs and milestone rewards for ~3K repeat buyers

3 Target Young-Adult Segment

Personalized promotions and digital campaigns for top revenue-contributing age group

4 Promote High-Rated Products

Use Boots and top-rated items in campaigns and cross-selling strategies

5 Increase Average Purchase Value

Product bundling, cross-selling, and threshold-based offers to raise the \$59.76 average

Future Scope

- Customer Lifetime Value (CLV) analysis
- Churn prediction modeling
- Advanced segmentation by frequency and spending
- Time-based seasonal trend analysis
- Automated Power BI data refresh
- Predictive product recommendation systems

Skills Applied

PYTHON & PANDAS

POSTGRESQL

SQL

POWER BI

DATA ANALYSIS

 **Project Completed** — End-to-End Data Analytics Workflow: Raw Data → Python & Pandas → Feature Engineering → PostgreSQL → SQL → Power BI → Business Insights → Recommendations